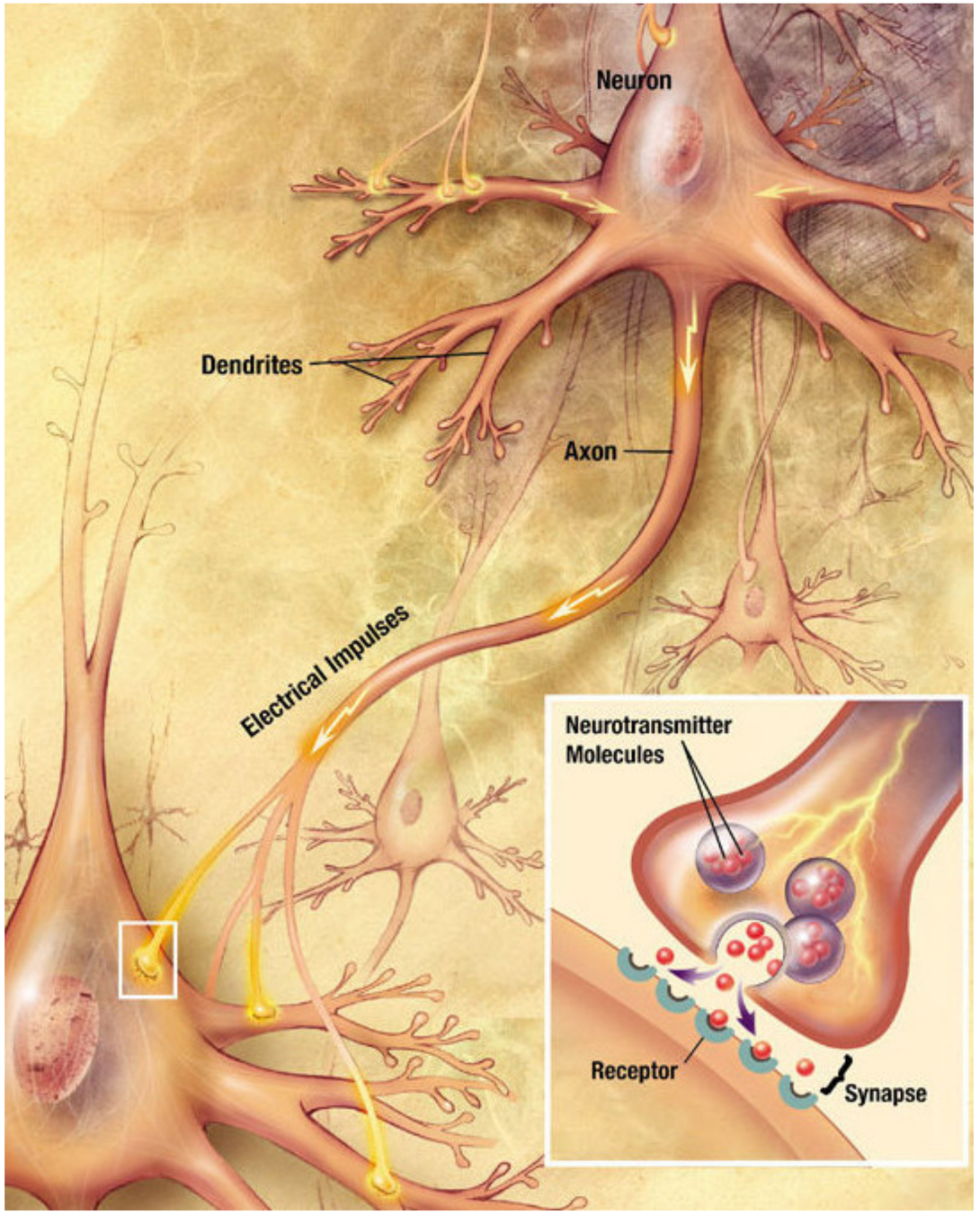
Neural Networks

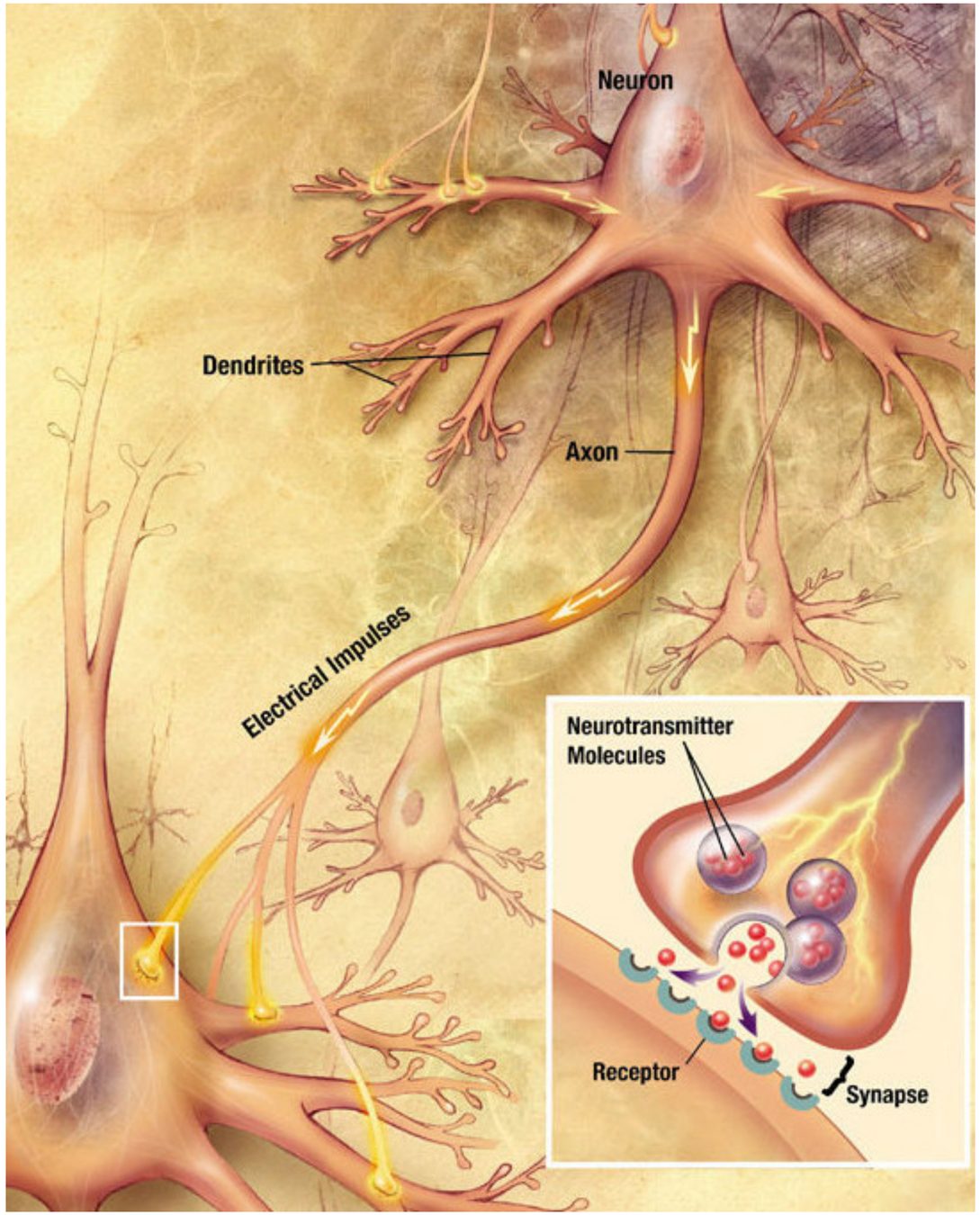
Machine Learning
CS4824/ECE4424
Bert Huang
Virginia Tech

Outline

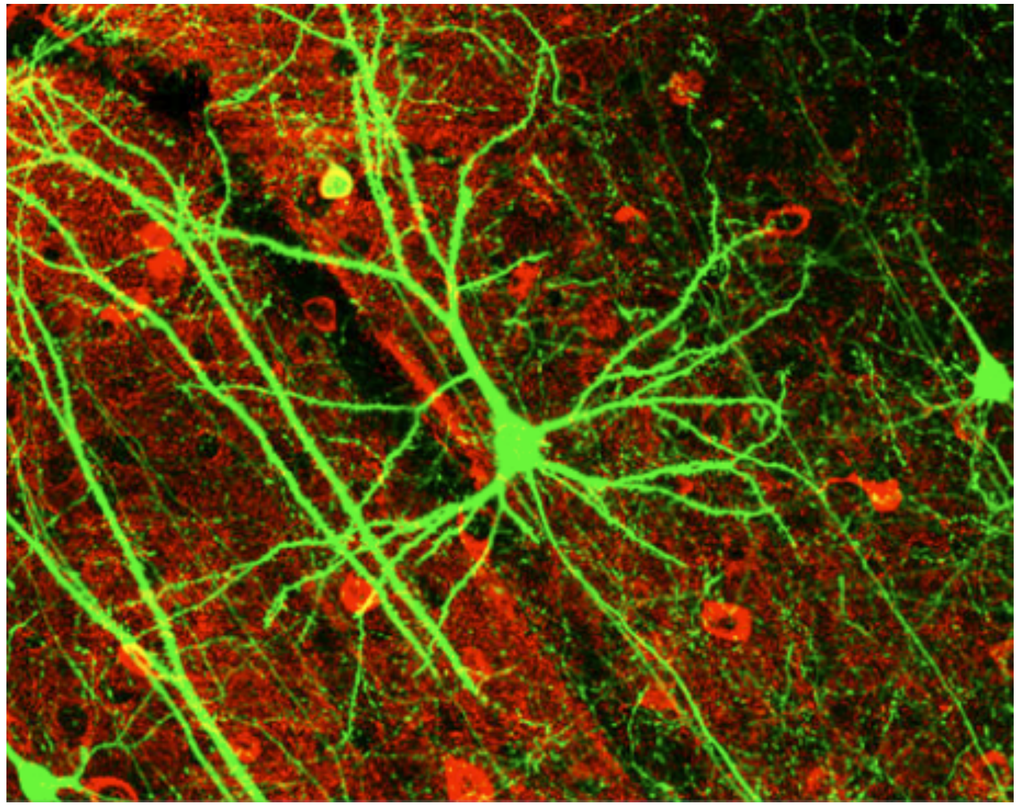
- Logistic regression and perceptron as neural networks
- Multi-layered perceptron
- Nonlinearity



https://en.wikipedia.org/wiki/Neuron#/media/File:Chemical_synapse_schema_cropped.jpg



https://en.wikipedia.org/wiki/Neuron#/media/File:Chemical_synapse_schema_cropped.jpg



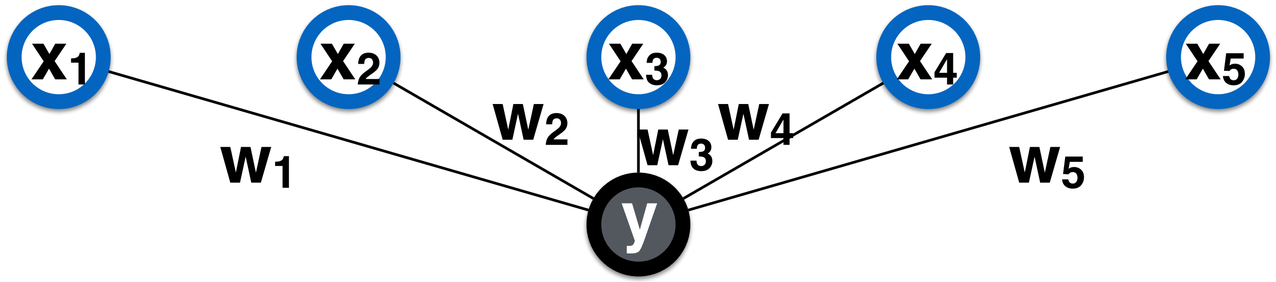
https://en.wikipedia.org/wiki/Neuron#/media/File:GFPneuron.png

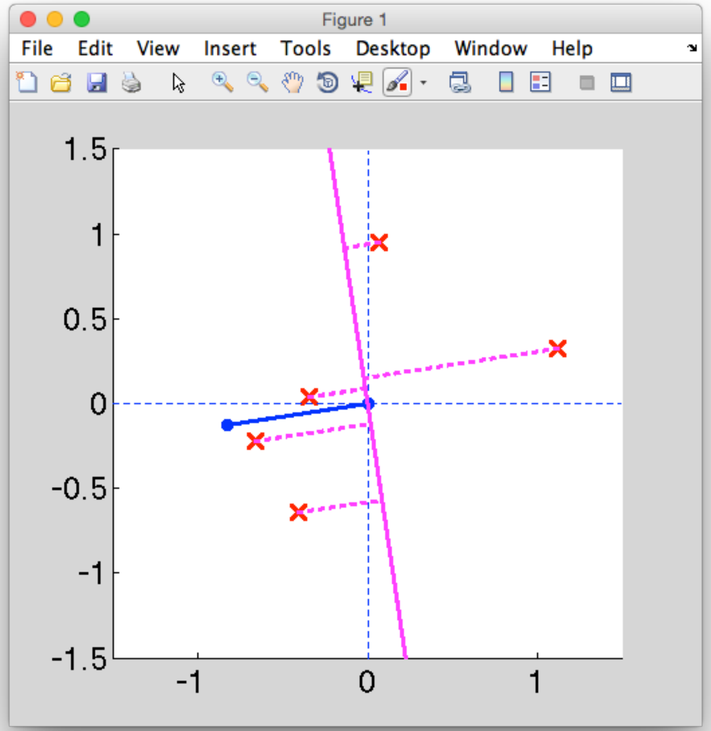
Parameterizing p(y|x)

$$p(y|x) := f$$

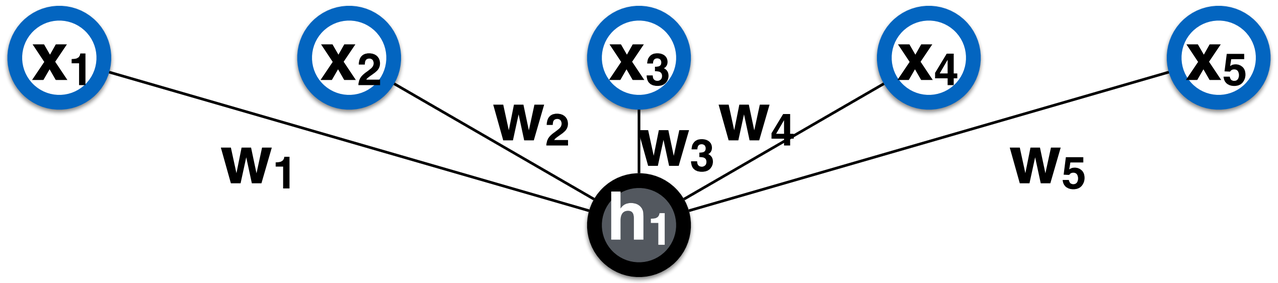
$$f : \mathbb{R}^d \rightarrow [0, 1]$$

$$f(x) := \frac{1}{1 + \exp(-w^\top x)}$$





Multi-Layered Perceptron

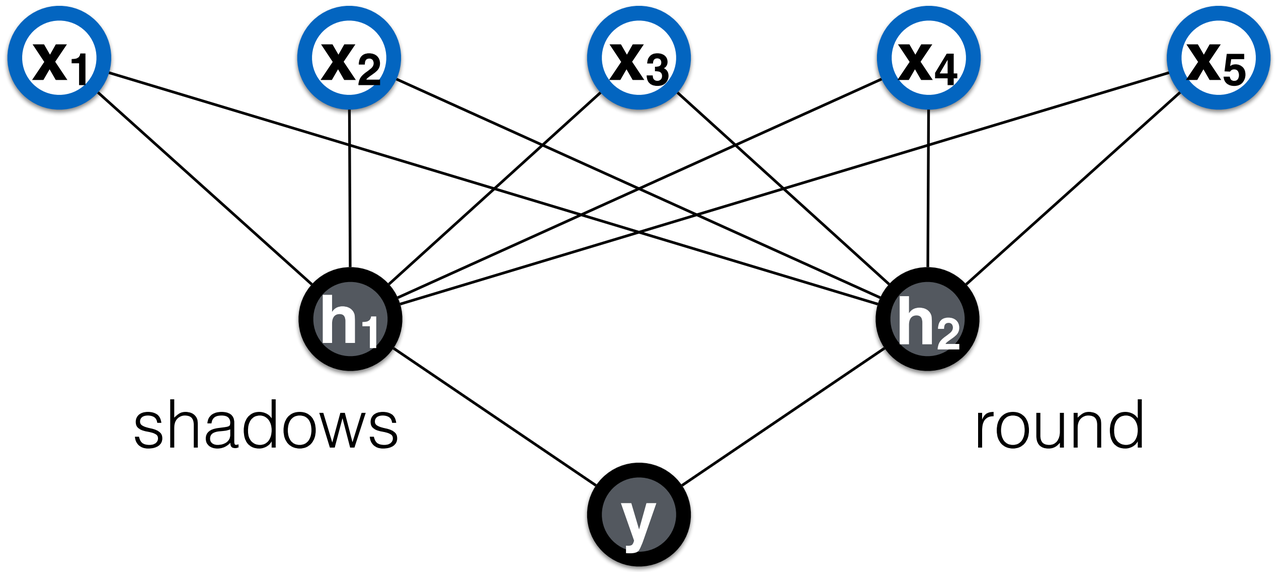


Multi-Layered Perceptron

raw data

representation

prediction

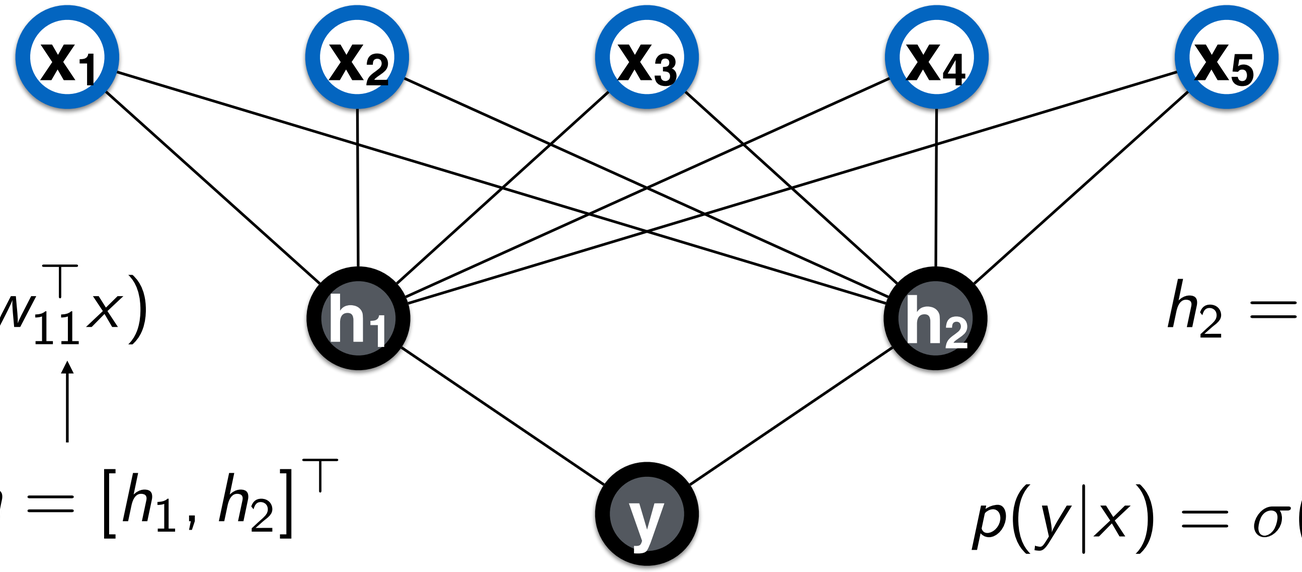


pixel values

shapes

faces

Multi-Layered Perceptron



$$h_1 = \sigma(w_{11}^\top x)$$

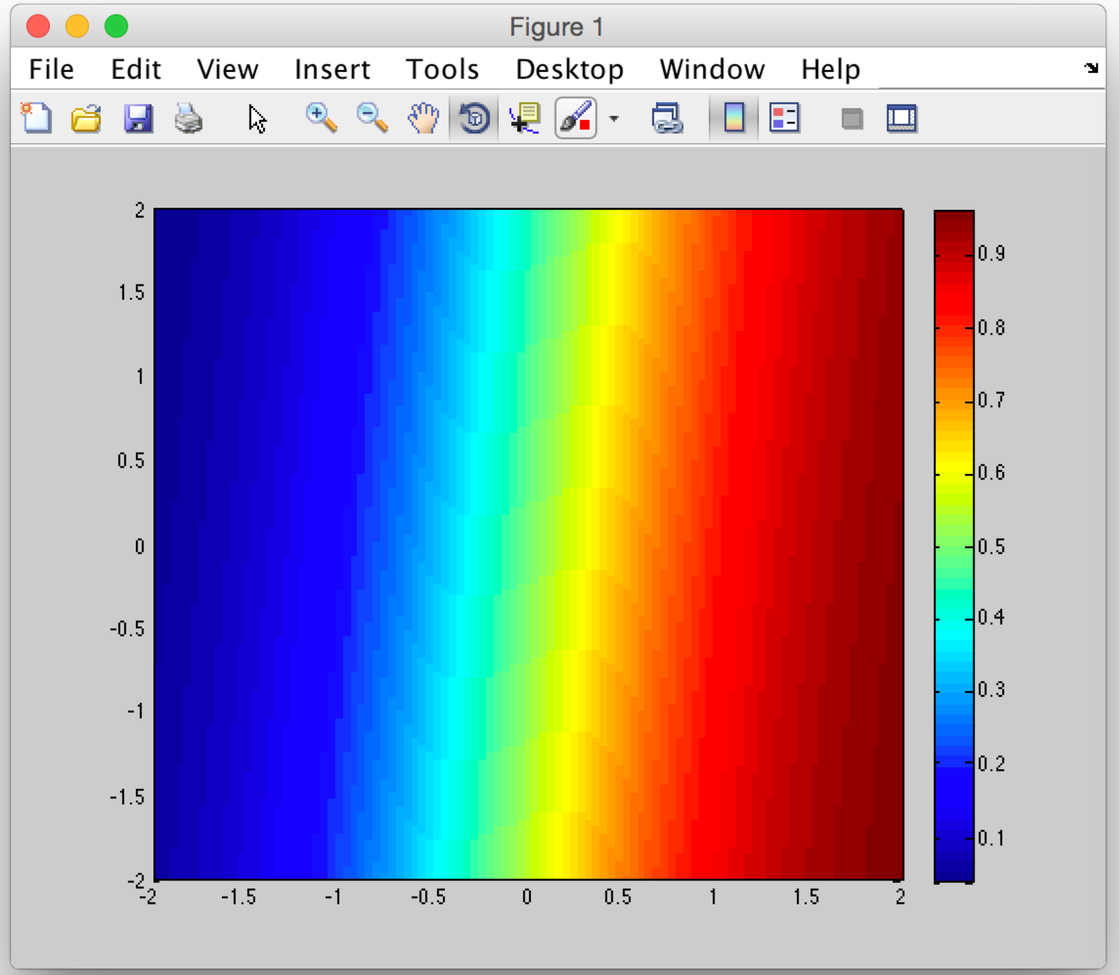
$$h = [h_1, h_2]^\top$$

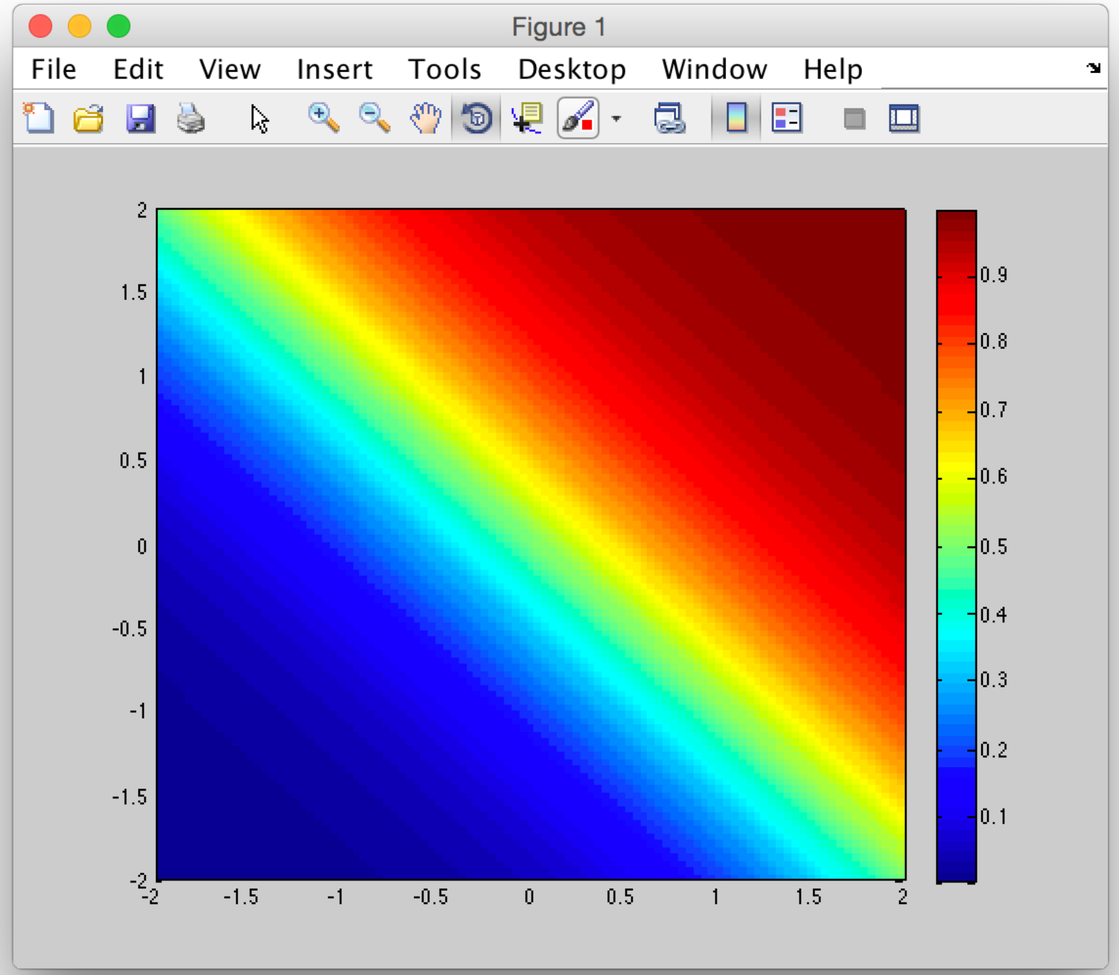
$$h_2 = \sigma(w_{12}^\top x)$$

$$p(y|x) = \sigma(w_{21}^\top h)$$

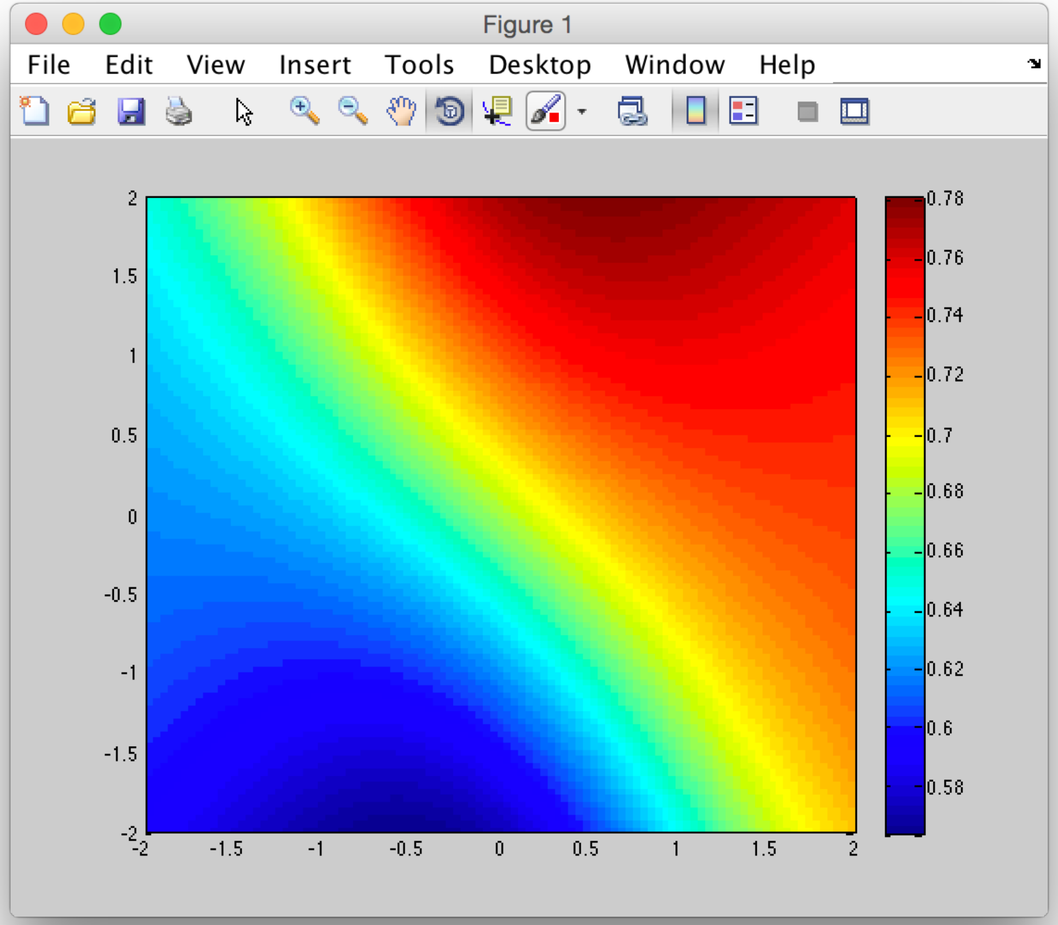
$$p(y|x) = \sigma\left(w_{21}^\top \left[\sigma(w_{11}^\top x), \sigma(w_{12}^\top x)\right]^\top\right)$$

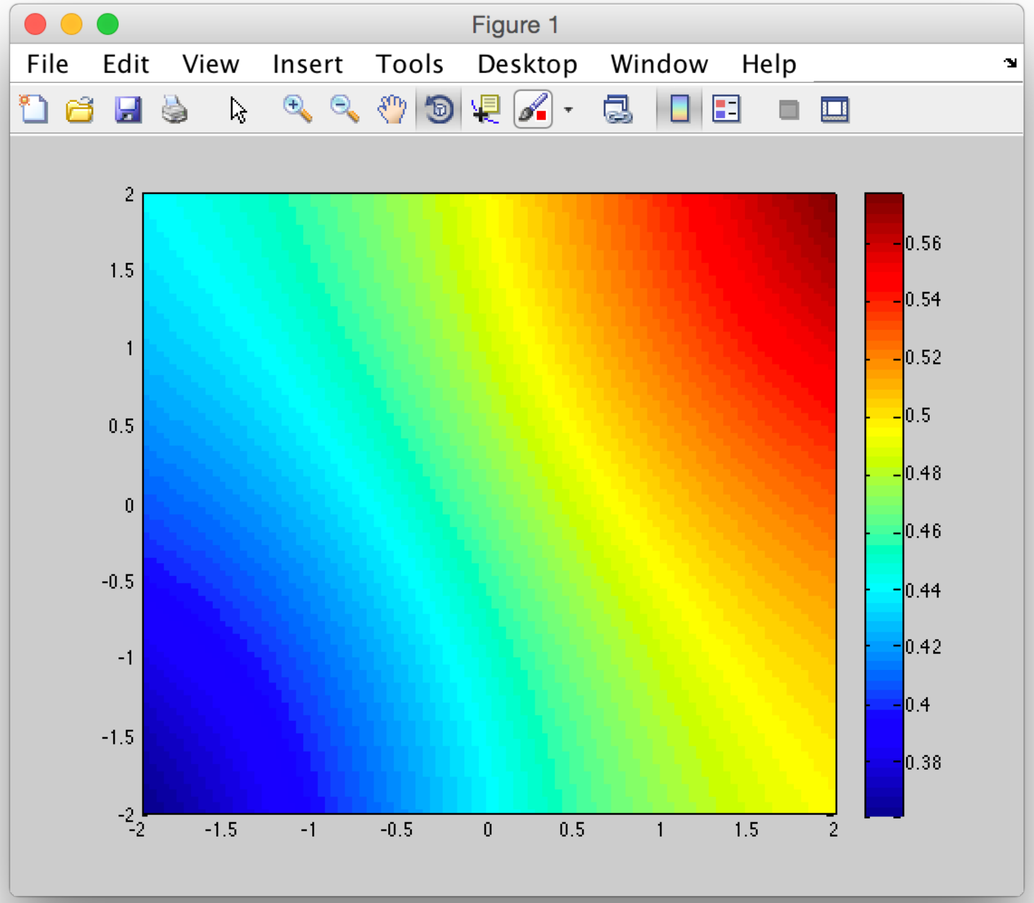
Decision Surface: Logistic Regression



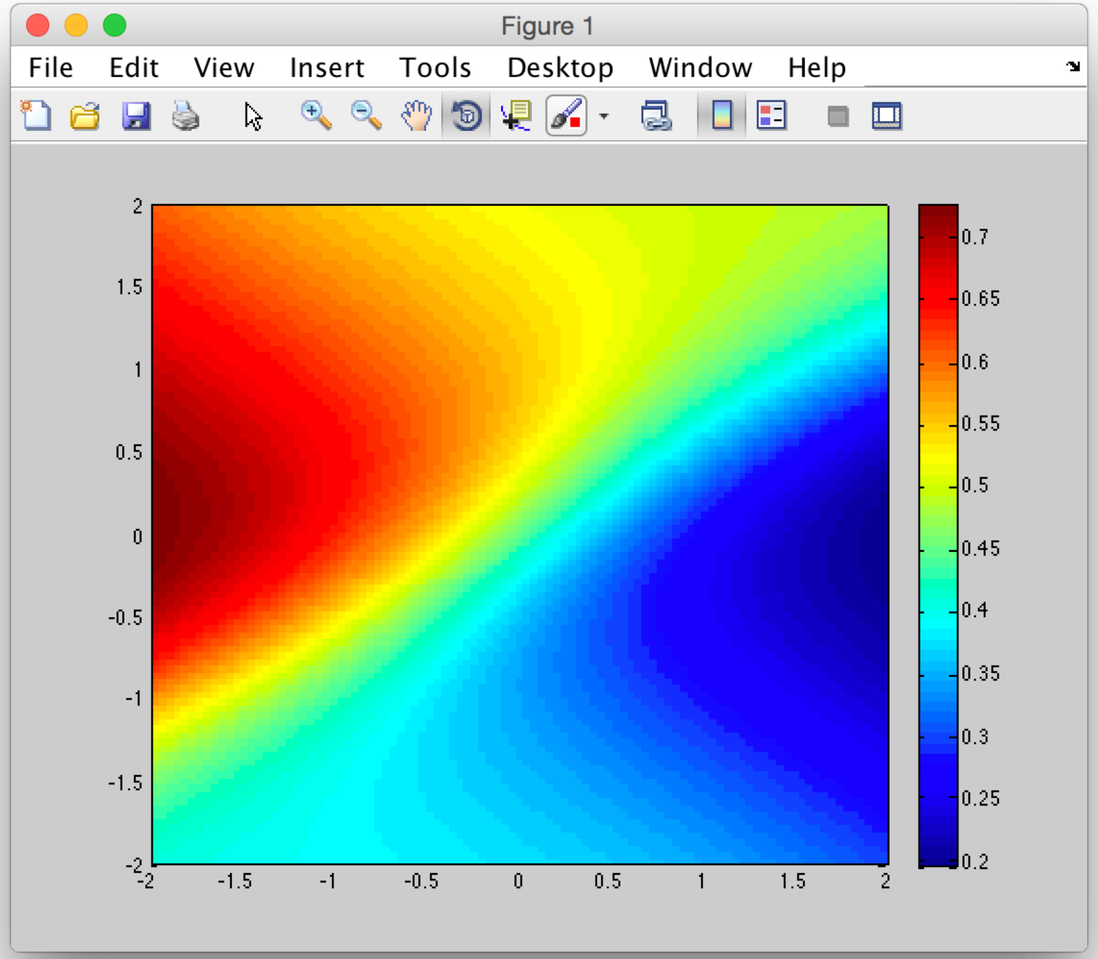


Decision Surface: 2-Layer, 2 Hidden Units

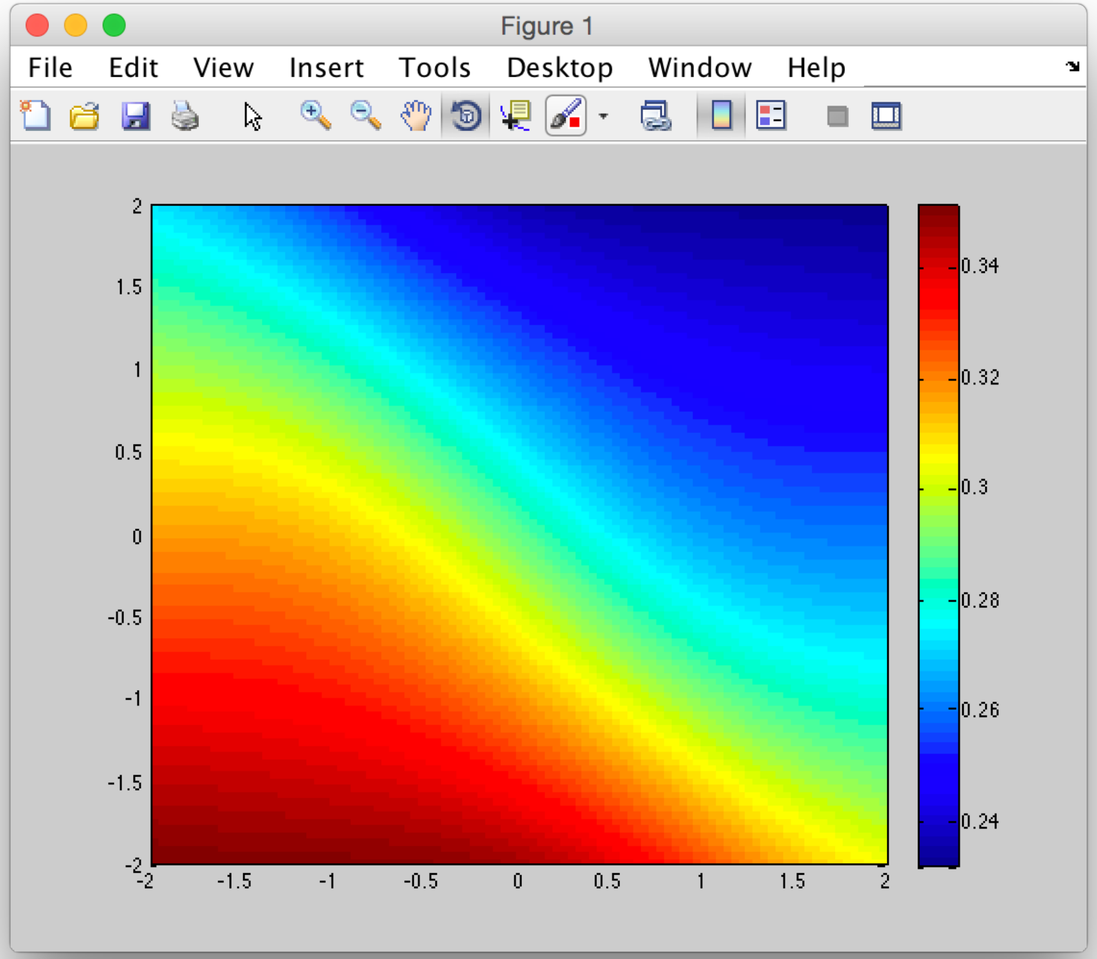




Decision Surface: 2-Layer, More Hidden Units

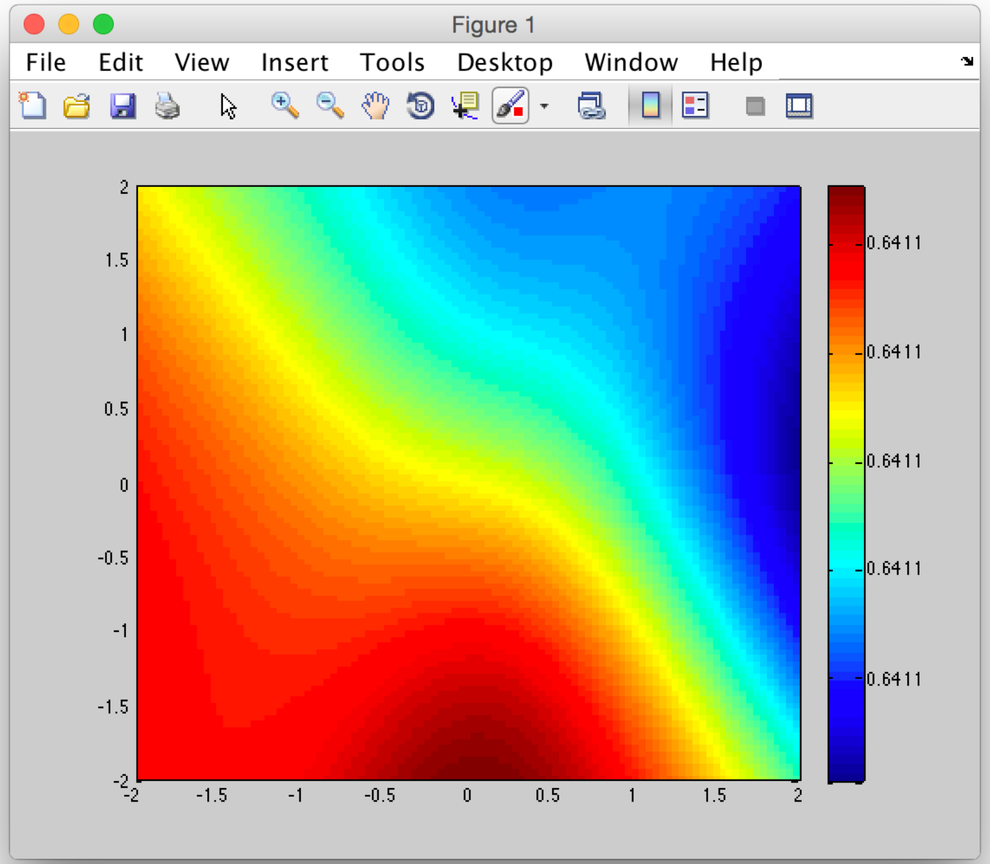


3 hidden units

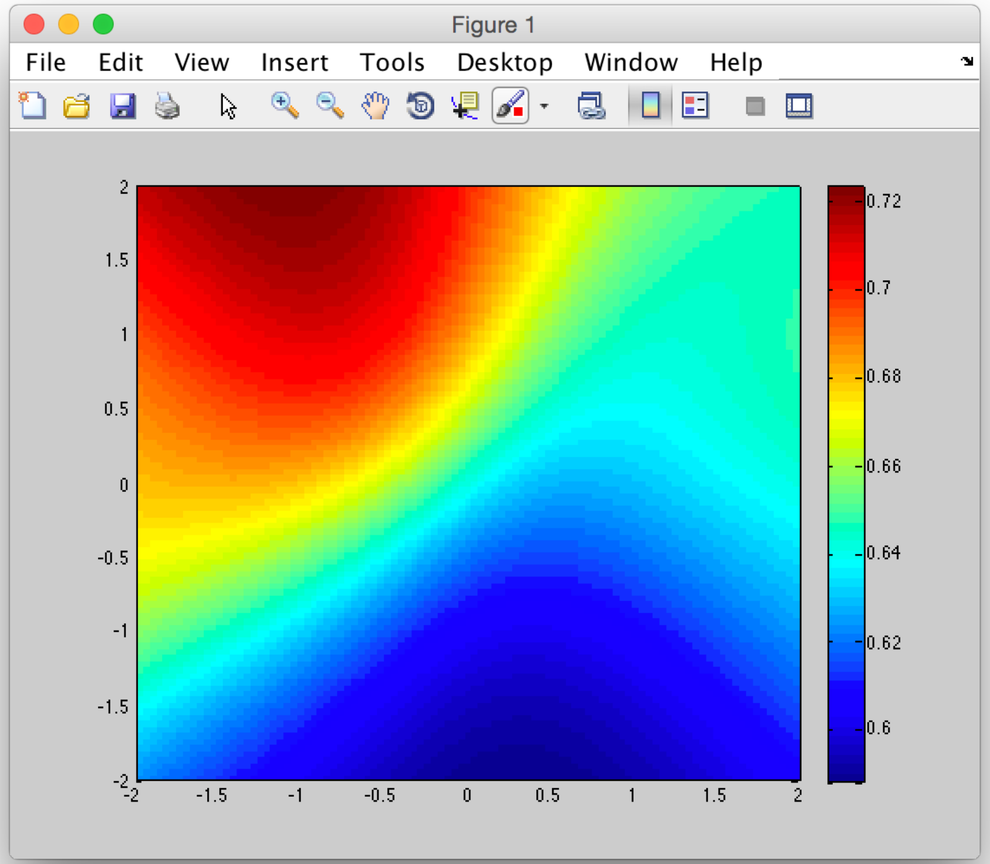


10 hidden units

Decision Surface: More Layers, More Hidden Units



10 layers, 5 hidden units per layer



4 layers, 10 hidden units each layer

Training

- Back propagation:
  - Compute hidden unit activations: **forward propagation**
  - Compute gradient at output layer: error
  - Propagate error back one layer at a time
- Chain rule via dynamic programming